



VOCATIONAL TRAINING REPORT

TRAINING TOPIC: ONGOING CONSTRUCTION SITE OBSERVATION

TRAINING LOCATION: NTPC BONGAIGAON BGTPP TOWNSHIP

TRAINING PERIOD: 45 DAYS

SUBMITTED BY: VAISHALI DHAR

BTECH CIVIL ENGINEERING AT NIT SILCHAR , 2ND YEAR

PROJECT: SHOPPING COMPLEX CONSTRUCTION

This page marks the beginning of my vocational training report, where I'll be sharing my observations and insights gained during my 45-day training period at the NTPC Bongaigaon BgTPP Township construction site.

SHOPPING MALL COMPLEX REPORT



PROJECT : SHOPPING MALL COMPLEX

LEAD BY : NTPC BONGAIGAON

REPORT SUBMITTED BY : VAISHALI DHAR

(REPORT FOR VOCATIONAL

INTERNSHIP)

CONTENTS

- 1. TERMS AND CONDITIONS OF THE CONSTRUCTION**
- 2. PROJECT OVERVIEW**
- 3. FOUNDATION, FOOTING AND COLUMN**
- 4. GENERAL REPORT**

TERMS AND CONDITIONS FOR CONSTRUCTION

1. Completion period of the project: 12 months.
2. Price variation: not allowed.
3. Payment Terms: (as per GCC civil works.)

Progress ("On-account"/RA) Payments *Clause 46 (On-account Payments):*

Payments certified by the Engineer-in-Charge are released periodically.

These are without prejudice to final account settlement.

Advance Payments *Clause 14.2 (Advance Mobilisation & Materials):*

Up to 10% mobilization advance (typically in two 5% tranches) against BGs.

Interest is charged (~8% p.a.) and recovered from RA Bills.

Security Deposit & Retention:

SD is deducted as per statutory rules.
A percentage of RA bills is retained by NTPC, released in two stages:

1.After Physical Completion

2.Upon Successful Completion of Defect Liability Period (DLP).

Final Bill & Time Limits

Clause 52 (Time Limit for Final Bill):

Final bill submission: within 3 months of physical completion.

Payment timelines:

≤ ₹5 lakh contracts: paid within 4 months

> ₹5 lakh contracts: paid within 6 months

Price Adjustment

Clause 53:

Base contract price is subject to price adjustment as per formula, unless otherwise stated.

SCC specifies the components and limits .

4. Defect Liability: (According to clause 45)

Standard DLP: Unless explicitly altered in the Special Conditions of Contract (SCC), the contractor is obligated to make good at their own expense any defects noticed within 12 months from the certified date of completion.

Latent Defects: For defects inherently present but not surfaceable during the DLP, the contractor's liability is extended to a period of up to 5 years from the end of the 12-month DLP.

-
5. **Construction Materials:** All construction materials are provided or arranged by the assigned contractor except the Fly Ash brick.
 6. **Transportation , Unloading, Loading to the construction site** is done by the agency at their own cost.
 7. **Fly Ash brick** is issued as Free Issue Material by NTPC.
 8. **Tools and Plants:** All tools and Plants are arranged by the contractor/Agency.
 9. **Agency to comply Labour law and to pay all the statutory benefit to workers deployed for the work.**
 10. **All works are executed as per FQP and CPWD / IS specification / BOQ and direction of Engineer in charge.**
 11. **Contractor is made responsible if their workers engaged in any unlawful activity or any unlawful happening.**
 12. **The Engineer In Charge is given the right to execute any work.**
 13. **The contractor or his representative should be present at the site daily.**
 14. **Agency will provide all type of safety appliances to their workers(safety shoe, gum boot, gloves, safety ribbon etc.)**
 15. **All debris/ waste materials generated during job is to disposed at NTPC scrap yard.**
 16. **The contractor to hire a Full Time supervisor Electrical & Civil works.**
 17. **Bidders to go through the meteorological condition of the working site before quoting rates**

18. Contractor to submit weekly progress report.

19. Agency to get CAR insurance before starting the work.

SERVICE DESCRIPTION

1. Excavation has been done by mechanical means as per DSR item 2.6.1
2. Excavation work for foundation work was done including the dressing of sides and ramming of bottom as per DSR item 2.8.1
3. Excavation work for pipes, cables including sockets plus dressing of sides and ramming of bottom was carried out for pipes and cables as per DSR item 2.10.1.1 and 2.10.1.2
4. Filling of available excavated earth was carried out as per DSR item 2.25
5. Surface dressing of the ground was done as per DSR item 2.28.1.
6. Clearing of grass, vegetation, jungle was done according to DSR item 2.31.
7. Chlorpyrifos/ Lindane emulsifiable concentrate of 20% was supplied as per DSR item 2.34.1.
8. Cement concrete for providing and laying in position provided as per DSR 4.1.8, 4.1.6, 4.1.3.

1:4:8 :8 graded stone aggregate 40 mm nominal size
1:3:6 :6 graded stone aggregate of 40 mm nominal size
1:2:4 :4 graded stone aggregate of 20mm nominal size
9. Centering and shuttering for foundations, footings, bases for columns as per DSR item 4.3.1

-
10. Damp proof of 40mm thick of cement concrete ratio 1:2:4 :4 graded stone aggregate 12.5 nominal size is provided
 11. A coat of petroleum bitumen layer of grade VG-10 is provided as per DSR item 4.13.
 12. Plinth protection of 50mm thick concrete of 1:3:6 :6 graded stone aggregate 20 mm nominal size was given as per DSR item 4.17
 13. Reinforced cement concrete of 1:1.5:3 (aggregate 20 nominal size) was provided up to plinth level as per DSR item 5.1.2.
 14. Reinforced cement concrete of 1:1.5:3 (aggregate of 20 nominal size) for walls, pilasters, buttresses, plinth and string courses, columns etc were provided as per DSR 5.2.2
 15. Centering, shuttering and removal of form work was done for foundations, footings, bases of columns as per DSR item 5.9.1
Walls including attached pilasters, plinth as per DSR item 5.9.2
Suspended floors, roofs, landings etc as per DSR item 5.9.3
Lintels, beams, plinth beams , cantilevers etc as per DSR item 5.9.5
Columns, pillars, posts as per DSR item 5.9.6
Stairs as per DSR item 5.9.7
Edges of slabs and breaks in floors and walls as per DSR item 5.9.16.1
Chajjas as per DSR 5.9.19
 16. Steel reinforcement for RCC work for under plinth and above plinth was provided as per DSR 5.22.6 and 5.22A.6

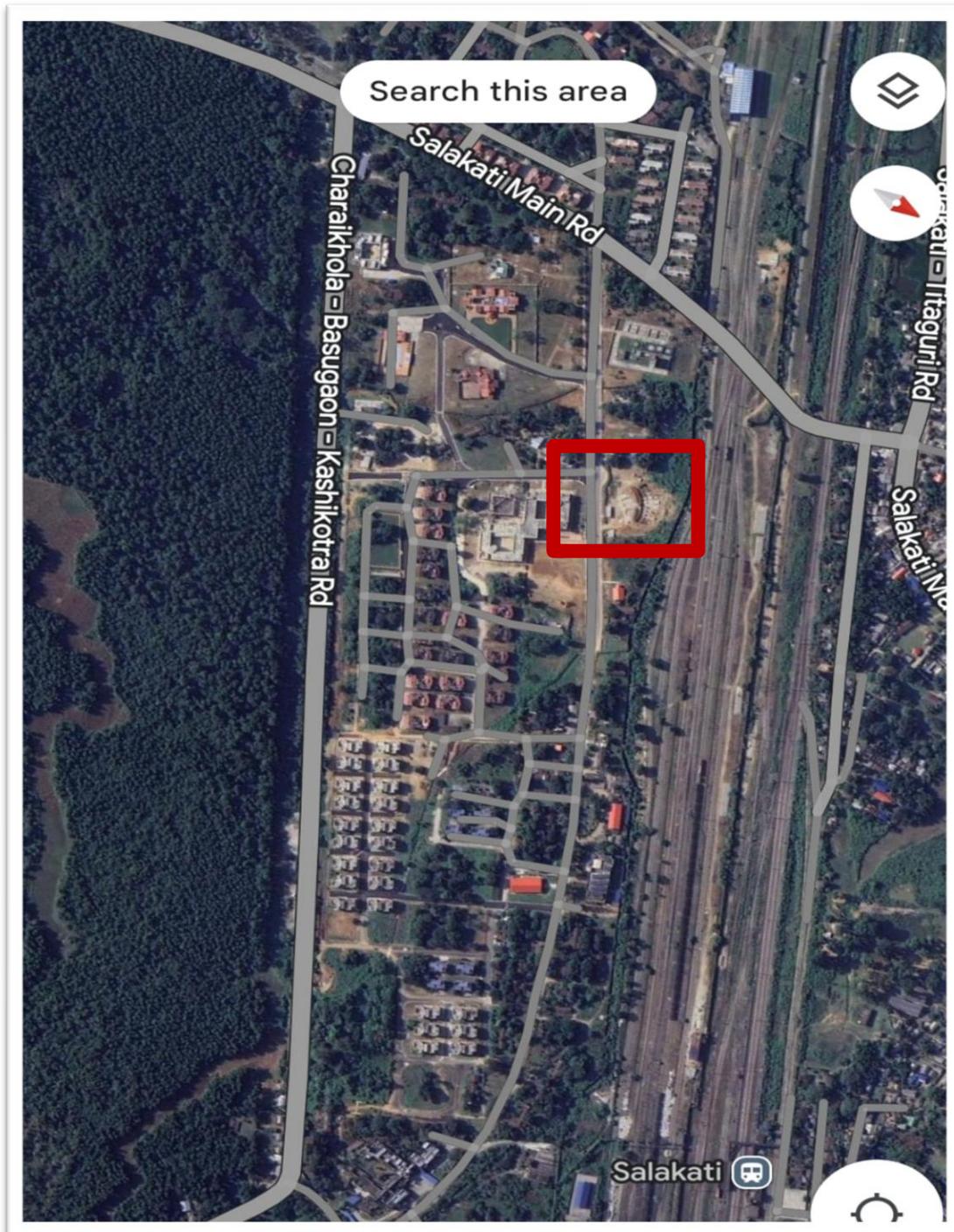
PROJECT OVERVIEW

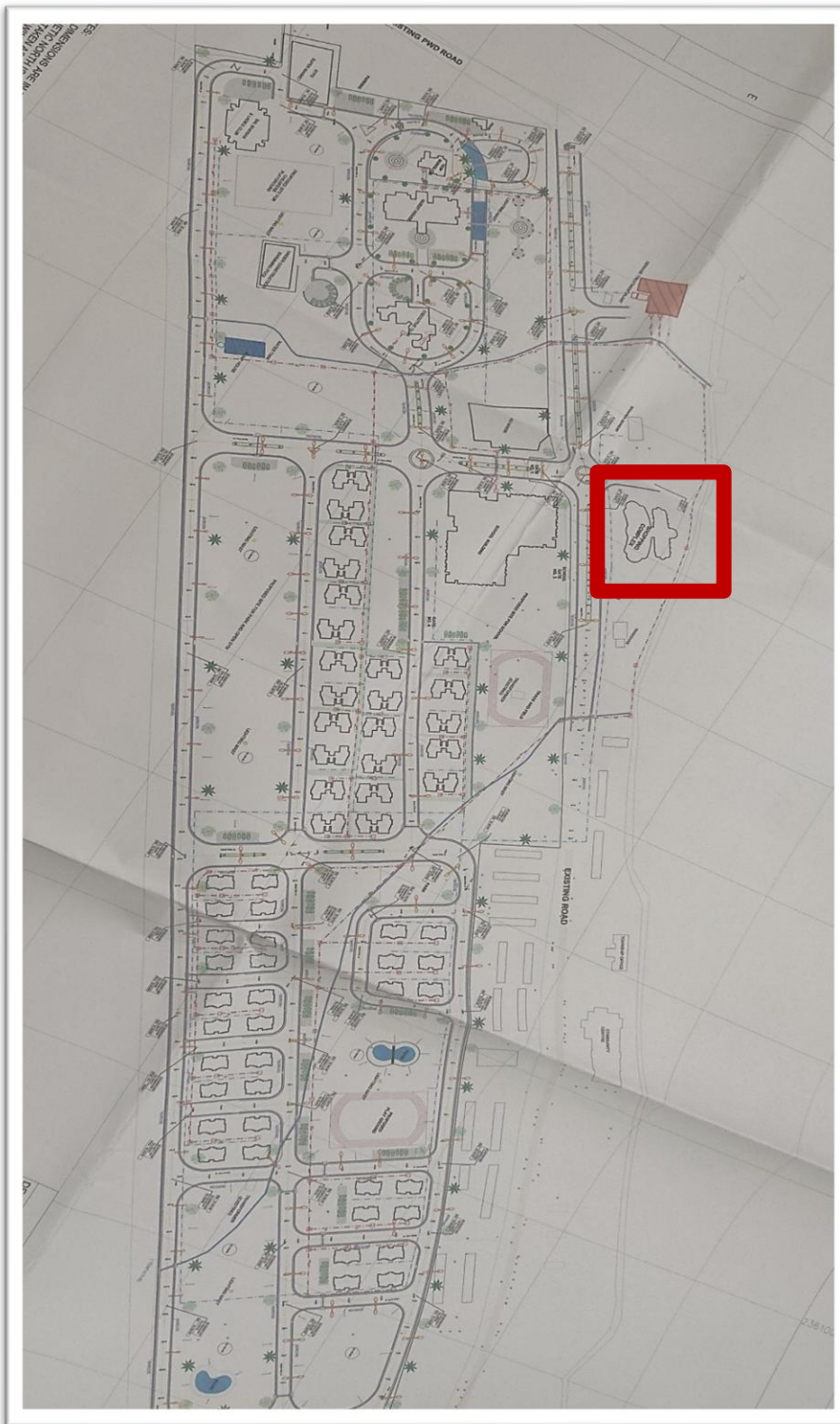
LOCATION OF THE SITE : coordinates : Latitude :26.437759

Longitude : 90.348137

Inside NTPC Bongaigaon BgTPP Township arena

Opposite of Norte Dame Academy.





PLOT AREA : 16m by 40m

BAMBOO SUPPLY :Charaikhola

TYPE OF CONSTRUCTION : Commercial Framed Structure where the beams ,columns , foundation and footing will take load with walls acting as cladding.

DESIGNED BY : Abhishek Kumar

CONTRACTOR'S NAME: Jehirul Islaam

AGENCY (FOR ESSENTIALS) : Two Star Enterprize

PROJECT START DATE : 14 Sept 2024

EXPECTED COMPLETION DATE : 12 months after the start date.

Description of the plan

1. Total 8 SHOPS :

Dimensions : 4000 mm * 5400 mm

Each provided with a Rolling Shutter of Dimensions (R.S) : 2400 mm *2700 mm.

2. ATM Room :

Dimensions : 3120 mm * 3000 mm

Provided with a Rolling Shutter of Dimensions (R.S) : 2400 mm * 2700 mm.sss

3. Electrical Room :

Dimensions : 4760 mm * 2700 mm

Provided with a Door (D1) of Dimensions : 900 mm * 2100 mm.

4. Store Room :

Dimensions : 4760 mm * 2700 mm

Provided with a door (D1) of Dimensions : 900 mm * 2100 mm.

5. Toilet (HE) :

Dimensions : 5145 mm * 2670 mm

Wash basin shelf of 900 mm

Provided with two Ventilators (V) : 600 mm * 750 mm

One Door (D2) of Dimensions : 750 mm * 2100 mm

One Window (W2) of Dimensions : 900 mm * 1200 mm

One Water Closet (W.C) of length 1600 mm

with a door (D2) of Dimensions : 750 mm * 2100 mm

6. Toilet (SHE) :

Dimensions : 2485 mm * 2670 mm

One Ventilation (V) of Dimensions : 600 mm * 750 mm

One Window (W2) of Dimensions : 900 mm * 1200 mm

One Door (D2) of Dimensions : 750 mm * 2100 mm

One Water Cabinet of length 1600 mm with a Door (D2) : 750 mm * 2100 mm

7. Toilet (PH) :

Dimensions : 2750 mm * 1800 mm

One Door (D) of Dimensions : 1000 mm *2100 mm

8. Stair case :

3 Steps of width 300 mm , 5545 mm length is given at the Entrance.

**12 steps of 300 mm width, 1500 mm length over a 3470 mm span each side.
Landing step of 1500 mm width leading up to First Floor.**

9. Lobby Passage :

1500 mm Width is given over both sides of the ATM Room.

10. Ramp:

On both sides at the Entrance of Dimensions : 3950 mm * 1500 mm.

11. Apart from that :

**Two Window (W1) of Dimension : 900 mm *2100 mm
each provided near Toilet(HE) and Toilet (SHE).**

**One window (W) of Dimension : 2000 mm * 1200 mm
Provided landing Platform of the Stair Case at the back.**

12.Floors : Ground Floor with Mumty level.

FOUNDATION FOOTING AND COLUMN

FOUNDATION WORK REPORT

PROJECT : Shopping Complex Construction

DRAWING NUMBER : 4610-508-TOC-C-101

TITLE : Foundation Layout Plan

DESIGNED BY : Abhishek Kumar

STATUS : Fully Executed

PROJECT OVERVIEW

1. All works in the drawing have been completed:
2. Site clearance and layout marking done with theodolite.
3. Excavation completed for all footing locations as per the grid up to 2m Depth.
4. PCC base of M-7.5 was laid with proper compaction.
5. Cement aggregate sand ratio of 1:4:8 for PCC
6. RCC footing cast using M25 concrete.
7. Cement aggregate sand ratio of 1:1:2 for Footing.
8. Reinforcement Fe500 placed and tied as per design schedule.
9. Vertical bars projected for future column works.
10. Curing maintained for more than 7 days post casting.

FOUNDATION DESIGN

The foundation design consists of isolated footings, combined footings, and a schedule of columns. The isolated footing schedule provides details on the size, bottom reinforcement, and other relevant information for each footing mark (F1 to F4). The combined footing schedule outlines the size, length, and bottom reinforcement, and other details for each combined footing mark (CF1).

SPECIFICATIONS

- Isolated Footings: Four isolated footings (F1 to F4) with the following details:

- F1: Size - 2600 mm x 2600 mm x 450 mm, Bottom Reinforcement - 12Y@150c/c both ways

- F2: Size - 2200 mm x 2200 mm x 450 mm, Bottom Reinforcement - 12Y@150c/c both ways

- F3: Size - 2000 mm x 2000 mm x 350 mm, Bottom Reinforcement - 12Y@180c/c both ways

- F4: Size - 1600 mm x 1600 mm x 300 mm, Bottom Reinforcement - 10Y@150c/c both ways

- Combined Footings: One combined footing (CF1) with the following details:
It includes C3, C4, C5 type columns.

- Size - 8781 mm x 3918 mm x 450 mm, Bottom Reinforcement :

Along-L/B (Bottom): 16Y@180c/c both ways

Along-L/B (Top): 12YU@180c/c both ways

- Columns: Five columns (C1 to C5) with different sizes and reinforcement details:

- C1: Size – 450 dia, Reinforcement : 8-16 mm dia bars, Stirrups : 2- 8 mm dia

- C2: Size - 300 mm x 600 mm, Reinforcement : 12-16mm dia bars, Stirrups:

2- 8mm dia.

- C3: Size - 300 mm x 600 mm, Reinforcement : 6-20 mm dia bars,6-16 mm Dia bars, Stirrups: 2-8mm dia

- C4: Size - 300 mm x 600 mm, Reinforcement :12-20 mm dia bars ,2-16 mm dia bars, Stirrups : 3-8 mm dia

- C5: Size - 300 mm x 300 mm, Reinforcement : 4-25 mm dia bars,4-20 mm Dia bars, Stirrups :1-8mm dia

RCC REQUIREMENT
FOR INDEPENDENT FOOTING

DIFFERENT FOOTINGS	LENGTH	WIDTH	DEPTH	PCC THICKNESS	RCC REQUIRED FOR PCC	RCC REQUIRED FOR FOOTING
F1	2.6 m	2.6 m	0.450 m	0.075m	0.5367 m3	3.042 m3
F2	2.2 m	2.2 m	0.400 m	0.075 m	0.388 m3	1.936 m3
F3	2 m	2m	0.350 m	0.075 m	0.322 m3	1.4 m3
F4	1.6 m	1.6 m	0.300 m	0.075 m	0.2104 m3	0.768 m3

No of Footings for each type :

F1 : 4

Total volume of RCC required for PCC : 2.1468 m³

Total volume of RCC required for Footing : 12.168 m³

F2 : 10

Total volume of RCC required for PCC : 3.88 m³

Total volume of RCC required for PCC : 19.36 m³

F3 : 8

Total volume of RCC required for PCC : 2.576 m³

Total volume of RCC required for PCC : 11.2 m³

F4 : 8

Total volume of RCC required for PCC : 1.6832 m³

Total volume of RCC required for PCC : 6.144 m³

REINFORCEMENT
REQUIRED FOR FOOTING

FOOTINGS	REINFORCEMENT DIAMETER	SPACING	NOMINAL COVER	TOTAL NO.OF BARS
F1	12 mm	150c/c	50mm	36
F2	12mm	150c/c	50mm	34

F3	12 mm	180c/c	50mm	26
F4	12 mm	150c/c	50mm	26

No of footings for each type:

F1 : 4

Total no of bars required : 144

F2 : 10

Total no of bars required : 340

F3 : 8

Total no of bars required : 208

F4 : 8

Total no of bars required : 208

RCC REQUIREMENT FOR COMBINED FOOTING

FOOTING	LENGTH	WIDTH	DEPTH	PCC THICKNESS	RCC REQUIRED FOR PCC	RCC REQUIRED FOR FOOTING
CF1	8.781 m	3.918 m	0.450 m	0.075 m	2.655 m3	15.478

Total 2 CF1 footings, total volume of RCC required : 30.956 m3

REINFORCEMENT REQUIRED
FOR COMBINED FOOTING

LAYER POSITIONS	DIRECTIONS	REINFORCEMENT DIAMETER	SPACING	QUANTITY
BOTTOM	ALONG LENGTH	16mm	180c/c	51
BOTTOM	ALONG BREADTH	16mm	180c/c	24
TOP	ALONG LENGTH	12mm	180c/c	51
TOP	ALONG BREADTH	12mm	180c/c	24

There are 2 CF1 footings so in total : $2 (51+24+51+24) = 2 (150) = 300$.

EXECUTION METHODOLOGY

1. **MARKING** : Grid layout with pegs and chalk line.
2. **EXCAVATION** : Manual and mechanical, checked by EIC.
3. **REINFORCEMENT** : Placed on cover blocks and tied.
4. **SHUTTERING** : Wooden/steel with bracing.

5. **CONCRETING:** Handled via mixers and compacted with vibrators.

6. **CURING :** Ponding and water sprinkling for 7–10 days.

CONCLUSION :

The foundation work has been completed in accordance with the approved design and specifications. The foundation plan and schedules provide a clear outline of the foundation elements, including isolated footings, combined footings, and columns. The special notes and general notes provide additional information on the design and construction requirements. Overall, the foundation work appears to be well-planned and executed.

GENERAL REPORT

-
1. All drawings were interpreted together with architectural, electrical and mechanical drawings
 2. Grid Dimensions were cross- checked and verified with architectural drawings before laying out.
 3. All dimensions were taken according to the architectural drawings.
 4. RCC footings were rest upto at least 0.5 m into hard rock.
 5. Undulated rocks were levelled accordingly to ensure stable and consistent footing foundation.
 6. Grade for columns , footings, beams and slabs etc were taken as M-25, as per IS 456.
 7. H.Y.S.D bars of Fe-550D and Fe-500D per IS 13920 ; grade Fe-415 per IS 1786 were used for reinforcements.
 8. All RCC works compiled with IS 456.

9. Nominal Cover to Reinforcement:

standard nominal cover requirements for various structural elements to ensure adequate protection of steel reinforcement from environmental effects. These values were followed as per IS 456 standards:

Structural Element	Nominal Cover
Slabs	20 mm
Plinth Beams	40 mm
Floor Beams	25 mm

Columns	40 mm
Columns (Below Plinth Level)	50 mm
Footings	50 mm
Rafts	50 mm
RCC Walls (Vertical Bar – Inner Face)	25 mm
RCC Walls (Earth/Water Face)	40 mm
RCC Walls (Dry Face)	25 mm
Stitch Slabs – Top Face	25 mm or bar diameter (whichever is greater)
Stitch Slabs – Bottom Face	40 mm

These specifications were implemented on-site and reviewed during quality checks.

10. Anchorage and Development Length

all beam bars were provided with anchorage lengths equivalent to the development length (L_d) plus 10 times the diameter of the bar (10ϕ). This anchorage was placed within the column at the beam end, as per standard detailing practices.

11. Distribution Bars

Wherever distribution bars were not specifically mentioned in the drawings, the default provision used was 8 mm diameter bars at 200 mm center-to-center spacing, as per standard engineering practice. This ensured proper load distribution and slab performance.

12. Placement of Slab Reinforcement

Reinforcement bars in slabs were placed such that bars along the shorter side were in the first (bottom) layer, while those along the longer side were placed in the second (top) layer. This method ensured proper structural behavior and reduced deflection.

13. Lap Length for RCC

For all reinforcement bars used in M-25 grade concrete, the lap length and development length followed was 50 times the bar diameter (50ϕ). This practice was essential for maintaining the continuity of reinforcement and minimizing structural weaknesses.

14. Column Lap Splicing Techniques

During the internship, I learned that lap splices in column reinforcement were only provided in the central half of the column height. I observed that hoops (stirrups) were placed over the entire splice length with a spacing not exceeding 100 mm center-to-center. I also got to know that no more than 50% of the bars were spliced at any one section to avoid weakening the structure.

15. Bar Lapping in Beams and Slabs

slabs and beams, bar laps were staggered and avoided at points of maximum bending moment. This detail was followed on-site to ensure the structural efficiency and durability of flexural members.

16. Use of Chairs and Layer Spacers

top reinforcement bars in slabs and rafts were supported using chairs (spacers made from reinforcement bars). These were provided at a minimum of 1 per square meter in areas requiring top reinforcement. I also noted that these chairs were tied to the main reinforcement and not placed directly on the shuttering, ensuring correct placement and cover.

17. Spacer Bars in Beams

spacer bars of 20 mm diameter at 800 mm spacing were used between two layers of reinforcement in beams. This helped maintain uniform separation between layers, particularly during concrete pouring.

18. Safe Formwork Removal

formwork removal was done with great care, especially to avoid vibrations that could damage green concrete. This practice was strictly followed on-site to preserve the quality and integrity of the cast elements.

19. Cantilever and Overhang Centering

centering under cantilevers, canopies, and overhangs was not removed until the required counterweight or back support had been established. This prevented structural imbalance and was considered a critical step in safe formwork removal. 24. Shuttering Removal:

The shuttering of cantilever slabs and beams was successfully removed, starting from the overhanging edges as per the specified procedure.

20. Utility Sleeves Installation:

The size and position of manholes, steel ladders, inlet/outlet pipes, and other services were executed in accordance with the plumbing drawings. All necessary sleeves were placed in slabs, beams, and walls before concreting.

20.Sunken Portions Casting:

All sunken portions of slabs and beams were cast monolithically with the respective structural elements, ensuring structural integrity

21.Chair Bar Specifications:

Chair bars for slabs were provided with a minimum diameter of 10 mm, while raft slab chairs were installed with a 20 mm diameter. Typical chair configurations were followed as shown in the drawing details

22.Necessary Embedments (Point 30)

Necessary embedments for staircase, and sailing were left before concreting, as specified in the design documents. This ensures that all structural elements are properly integrated and secured.

23.Filling Work :

Filling work for the building, plinth, and roads was carried out in accordance with the Bill of Quantities (BOQ) and Specifications for Execution of Structural Engineering Services (SPEES). This guarantees that the filling work meets the required standards and specifications.

24.Reinforcement Steel :

All reinforcement steel was bent and fixed in position as per IS:2502, ensuring that the steel reinforcement meets the required standards for strength and durability.

25.CI/RWP Installation :

No Cast Iron (CI) or Reinforced Wire Pipe (RWP) was allowed to cross beams or columns in plan unless noted on structural drawings. This ensures that the structural integrity of the building is maintained.

26.Reinforcement Detailing :

Detailing of reinforcement was carried out as per IS:13920, which provides guidelines for the detailing of reinforcement in concrete structures.

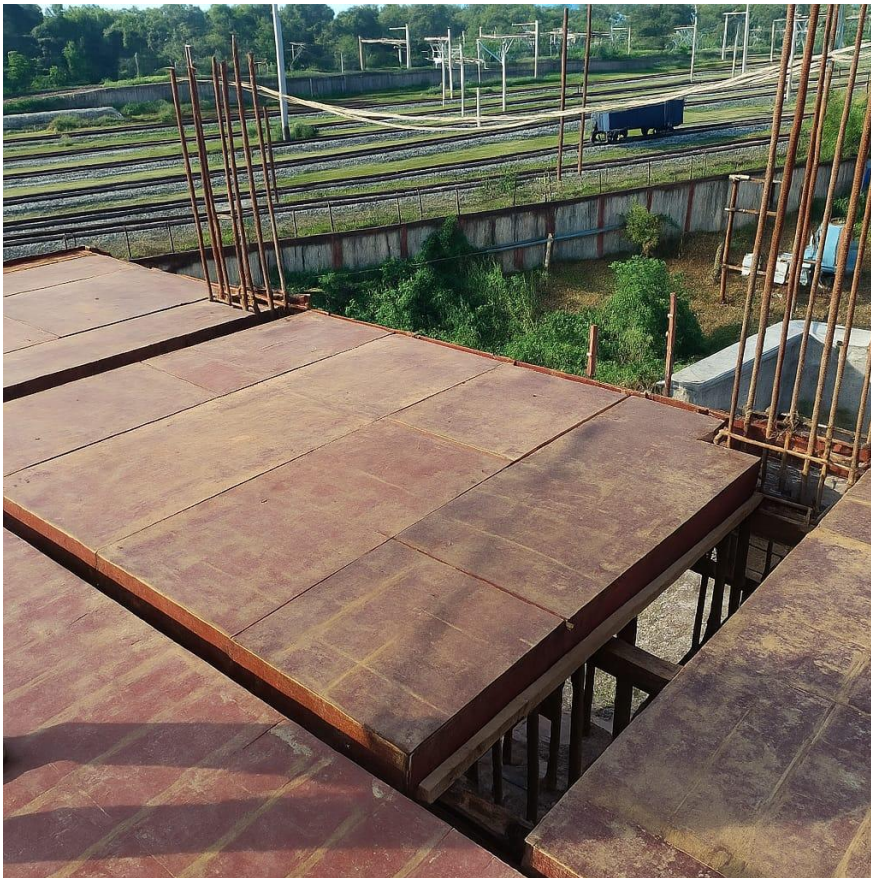
27.Longitudinal Bars in Beams:

In beams, longitudinal bars were spliced only if hoops were provided over the entire splice length, at a spacing not exceeding 150mm. The lap length was not less than the bar development length. Lap splices were not provided within a joint, within a distance of $2d$ from the joint face, or within a quarter length of the member. Not more than 50% of the bars were spliced at one section/location.

SITE VISIT DAY 1

dated : 26.06.2025

- 1. During our site visit, we observed on going construction activities at the first floor level, where slab work was in progress. Mainly shuttering work was in process along with filling of the frame work. Basically it is that area which is being prepared for installation of water tank.**



- 2. Plastering work was actively underway :**

6mm plaster was being applied to the ceiling and bottom of chajjas.

12mm plaster was being applied to the inner surface walls, top surface of chajjas.

The plastered walls were being curved with water to allow the mortar to gain proper strength and bonding.

3. The following structural elements has already been completed :
(ground floor + roof(first floor level))

Footings

Columns

Beams

Plinth beams installation

Slab work for roof

Staircase at the back side.

Inner walls

Parapet walls

Yet to be completed as observed :

Entrance Ramp



Entrance Staircase

Ground Slab

Slab work for the area installed for water tank.

Windows, Doors installation

Finishing and flooring

- 4. Pre support structures were made near back staircase to support the roof shuttering work of stair cabin.**



5. Electrical installation:

The electrical conduits were being fitted by first clearing the applied concrete layer (i.e for creating channels), followed by placing the electrical pipes.



Covering the walls along with pipes with chicken mesh wire.



Applying a thin mortar layer over the mesh.

Final finishing layer to be done after this preparation.

An electrical control room is located behind Shop 4, which serves as the main source of power distribution.

From this room :

Electricity is routed to distributor panels located within Shop 4 and Shop 5.

Power is further distributed to individual rooms and toilets via concealed wiring.

- 6. Water tank to be made of 150 mm thickness of M25 concrete , 3 m x 3m x1 m with slab thickness 125 mm**

SITE VISIT DAY2

Dated 30.06.2025

1. Plastering Work

Finishing work was actively ongoing at the site.



12 mm plastering was carried out on the inner surfaces of the walls in the Lobby area.

Plastering of shop walls was nearly complete, except for the ATM booth and a few other remaining rooms.



Curing of plastered surfaces was being done to ensure proper bonding and strength.

2. Shuttering and Roof Reinforcement

Shuttering work was completed for the stair cabin roof, which also serves as the support structure for a 150 mm thick water tank (dimensions: 3 m × 3 m × 1 m).



Shuttering was done using 8 ft × 4 ft plywood sheets.

Reinforcement bars were placed and tied with stirrups at 75 mm and 100 mm gaps in the beam sections.

Nominal concrete cover was maintained as per instructions.



3. Support and Pre-stage Work

Pre-stage support systems were being installed to assist with structural stability and to facilitate worker movement on-site.



4. Structural and Masonry Work

Flooring has not been done.



Brick masonry for the entrance staircase was being executed.



Walls near the rear staircase windows were still under construction.

Ramp work had not been started as of the reporting date.



5. Pending Works

Floor finishing is pending.

Installation of doors and windows is yet to begin.

7. ~4000 bags of Cement till were used.

8. ~80 metric Ton steel used.

